

Kathmandu Metropolitan City

Education Management ERP Project Document

Stage One – Government Schools

1. Introduction

Kathmandu Metropolitan City is the largest local government in Nepal. It is responsible for providing, supervising, and supporting education within its boundaries. Within its Metropolitan boundary operate ninety government schools along with special schools, and more than five hundred and thirty private schools. The Education Department of the Metropolitan City performs a wide range of tasks such as school registration, teacher quota management, payroll, infrastructure development, monitoring of learning environments, and coordination with the Ministry of Education and other agencies. Although these activities have expanded over the years, the supporting administrative systems have remained mostly paper-based or divided across spreadsheets, databases, and separate applications. Because of this situation, it is difficult to maintain a single accurate version of data, to plan investments effectively, or to monitor results in real time.

To solve these problems, the Metropolitan City plans to develop and implement an Education Management Enterprise Resource Planning system. The ERP will work as a single digital platform where all information about schools, teachers, students, budgets, and resources can be stored, updated, and analyzed. The first stage will focus on government schools and the special school so that the city can build a strong foundation before including the private schools in the second stage.

2. Vision and Guiding Principles

The guiding vision of the ERP project is to create a digital ecosystem that connects every school to the Metropolitan City through a transparent, data-driven, and service-oriented approach. The system will turn routine administrative work into a continuous flow of valuable information that supports better decisions for teachers, parents, and policy makers.

The design will follow four principles. First, all data must belong to the city and remain protected but easily accessible to authorized officials. Second, the system must interlink with national and provincial systems rather than replacing them. Third, every process must eventually be available online, mobile, or offline so that no school is left out because of connectivity issues. Fourth, technology should reduce effort and not add complexity, therefore forms must be simple and workflows should match existing administrative practices.

3. Core Objectives

The project has multiple levels of objectives. At the operational level it seeks to automate the daily activities of schools and the education department. At the managerial level it provides comprehensive monitoring and financial control. At the strategic level it facilitates policy analysis, forecasting, and continuous improvement. The objectives are to create a digital profile

for every school, to automate the payroll system with full financial integration, to link all sources of investments and budgets with physical results in schools, to improve teacher management and performance records, to connect with the national EMIS and provincial information systems, and to prepare the data foundation for student-level applications such as learning analytics in future stages.

4. Project Scope and Coverage

Stage one covers all government and special schools located inside the thirty-two wards of Kathmandu Metropolitan City. The private schools are included in the database structure and will be activated in stage two. The scope extends beyond simple record keeping to include intelligent analytics, performance dashboards, digital communication tools, and mobile accessibility. The system will be prepared for multilingual operation, currently supporting English and Nepali with room for additional languages if required later.

5. Functional Architecture

5.1 School Information and Profile Module

Each school will have a living digital identity that includes its name, code, location, land record, buildings and classrooms, laboratories, libraries, and condition of facilities. It will show energy sources, digital connectivity, attendance of students, academic streams, and details of inclusive features such as ramps, accessible toilets, and special education resources.

The profile will also maintain the names of principals, management committee members, school improvement plans, disaster preparedness records, and links to photographs, maps, and design documents. Every update made by the school or ward will create an automatic time-stamped history. The module will therefore serve as the single reference for infrastructure planning, policy compliance, and asset management.

5.2 Teacher and Staff Management

The staffing module will handle every stage of a teacher's career within the Metropolitan City. It will store appointments, promotions, transfers, training history, performance evaluations, and professional development logs. The system will separate centrally sanctioned positions (Darbandi) from positions funded by the metropolitan city. It will be able to match available teachers with actual classroom requirements and will generate alerts when vacancies remain open for a long time or when surplus staff exist in certain schools.

A digital teacher file will contain scanned copies of educational certificates, citizenship records, service history, and disciplinary letters. Ward officers and city administrators will be able to view these files immediately instead of searching through paper archives. The module will later connect to nationwide teacher licensing and transfer systems when the national portal becomes available.

5.3 Payroll and Financial Management

The payroll module of the ERP will handle salary processing automatically. Salary scales, grade levels, steps, allowances, and deductions will be entered once according to the government pay structure, and the system will then use these details every month to calculate pay for all teachers and staff. Each school will confirm attendance through manual entry or simple biometric devices before salary generation so that payment is based on verified presence. The system will calculate salary amounts, including basic pay, allowances, provident fund, gratuity, tax, and other deductions, and will prepare pay slips automatically. School head teacher and account officers will review the calculations, and the Education Department and Finance Division will provide final authorization. Once approved, payment data will move electronically to the Metropolitan Education department for direct deposit into employees' bank accounts. All digital pay slips and annual statements will be stored in the ERP for staff reference, audit, and pension processing.

Because salaries for many teachers are funded fully or partly by the Metropolitan City, the ERP will record the source of funds for each payment so that reports clearly show how much is paid from the central government and how much from KMC resources. The system will not allow payments that exceed sanctioned quota positions and will keep a complete audit trail of every transaction. It will also provide forecasting tools that analyse salary scales, promotions, and grade increments to show expected financial needs for upcoming fiscal years. This predictive budgeting capacity will help KMC plan payroll budgets and manage cash flow based on accurate information rather than estimation.

5.4 Budget, Investment, and Resource Tracking

The resource tracking module will follow the complete lifecycle of every financial investment. Budgets will be created at the metropolitan and ward levels and broken down by program category. Each school will have its annual allocation visible in the ERP. When money is released, the school will receive a digital authorization and record actual expenditures through an easy interface. Invoices, photographs, and progress reports can be uploaded from mobile devices. Projects funded by other government programs, foreign aid, or social organizations will require preliminary approval through the system before implementation. This approval process will prevent duplication of efforts and ensure that all funds flow through recognized channels. The system will also include an inventory section that lists physical assets such as furniture, laboratory equipment, computers, and teaching materials. When new items are purchased their serial numbers and photographs can be recorded so that an accurate inventory is always available.

5.5 Integration with EMIS and External Systems

The ERP will exchange data regularly with the national EMIS through a secure Application Programming Interface if possible otherwise can be imported after exporting from EMIS system. The first integration will pull existing school codes, enrollment figures, and teacher quotas. Later it can send updated data back so that both systems show consistent results.

5.6 Attendance, Examination, and Learning Analytics (Future Ready Features)

Although stage one concentrates on administrative and financial aspects, the database will be designed to hold analytical information for later modules. This includes student attendance totals, internal examination results, and simple learning indicators. Once these modules are activated, the education department will be able to identify improvement trends and link them to investments, for example demonstrating whether digital classroom programs have led to better science results.

5.7 Communication and Citizen Engagement

The system will include message and notification tools so that KMC can send circulars or emergency alerts to all schools. In a later phase, parents may receive information about school activities, admissions, or events through a public portal. This feature will gradually strengthen community participation and transparency.

5.8 Book Free Friday and Vocational Learning

The ERP will include a module to support the Metropolitan City's **Book Free Friday** program, a weekly initiative that replaces traditional textbook lessons with practical and creative learning. Every Friday, students from Grade 1 to 10 take part in vocational and innovation activities designed to build life skills and confidence. Through this module, each school will prepare an annual plan that lists the activities or clubs available for students, such as gardening, art, music, computer use, basic engineering, health care, or environmental protection. The plan will show which teachers are responsible for guiding each group and how many students are enrolled. The system will keep these records and allow updates at any time during the year. On Fridays, schools will record attendance, note what activities were carried out, and upload photographs or short summaries of student work. This information will be visible to ward offices and the KMC Education Department for review and feedback. Any materials or funds used for Book Free Friday will automatically appear in the resource-tracking part of the ERP so that the city can see how its support has been used. Data from this module will link directly to student and teacher profiles. For students, participation will become part of their co-curricular record showing interests and practical skills. For teachers, it will count as innovative teaching activity. The ERP will generate simple reports showing how many schools run the program, how many students take part, which activities are most popular, and how the initiative is improving learning opportunities.

This module makes Book Free Friday organized and measurable inside the ERP. It helps every school plan, record, and share its vocational learning work and allows the Metropolitan City to promote creative, skill-based education across all government schools.

6. Workflow and Data Movement

The workflow is designed to mirror natural administrative hierarchies. Schools will enter and verify their own data. Ward education sections will check and approve entries. The metropolitan Education Department will review and finalize data, which then becomes available to the Finance and Planning Divisions. Audit logs will record each step with the name and date of the responsible user. Payroll runs will proceed in monthly cycles. Schools confirm attendance, ward verifies, and KMC finance executes payment. Resource investments will

follow yearly project cycles, beginning with proposal submission, then approval, fund release, purchase, verification, and closure. Feedback loops ensure that information about executed projects updates infrastructure records in real time. Data from EMIS will enter the system through scheduled synchronization, and a reporting engine will automatically compile tables for the annual education census.

7. Reports and Analytical Dashboards

The reporting engine will produce descriptive, analytical, and predictive views. Every user will have role-appropriate dashboards. Principals can see student and teacher data, ward officers can see all schools within their area, and senior officers can see city-wide performance. Typical reports will include enrollment summaries, staffing balance, payroll expenditure trends, investment utilization rates, and lists of schools needing infrastructure improvement.

In the advanced stage, visual dashboards will use graphs and maps to show data geographically. Decision makers can click on any ward to view the latest status of schools, projects, and expenditures. Export formats will include Word, Excel, and PDF, ready for submission to higher authorities.

8. System Features and Standards

The entire system will operate through a secure web platform accessible by computers and mobile devices. The design will follow responsive standards so that screens display comfortably on tablets or smartphones. Offline data entry options will allow schools without continuous connectivity to record data that synchronizes later.

All interfaces will be bilingual with Nepali as the primary language. Data storage and hosting will take place inside Nepal in compliance with national ICT policies. Encryption and role-based access will protect sensitive information such as salary or personal details. Automatic backup, disaster recovery arrangements, and multi-factor authentication will ensure reliability. The system will support upward scalability to more than one thousand institutions.

9. Governance and Data Ownership

Data ownership rests entirely with Kathmandu Metropolitan City. Policies will define who can view, edit, approve, or export information. The Education Department will manage user rights while technical administration remains under the city's IT division. Periodic data quality checks will be mandatory. The system will record every change so that audit teams can recreate any version of data if needed. These records will strengthen public accountability and legal compliance.

10. Sustainability and Future Expansion

The ERP has been planned as a long-term platform rather than a short-term project. Its modular structure allows new functions to be added without rewriting existing ones. Future modules already envisioned include private school regulation, student records from early childhood to secondary level, examination management, e-learning content distribution, attendance tracking

through mobile apps, transportation management, asset depreciation, and integration with citizen grievance systems.

Eventually, the platform can connect to national digital identity or payment services, allowing online fee payments and certificate verification. The system can also host advanced analytics based on artificial intelligence, such as predicting dropout risk or optimizing resource allocation.

11. Expected Outcomes and Impact

After implementation, the Metropolitan City will experience a transformation in how education data is managed. Information that previously required weeks to collect will be available instantly. Expenditure tracking will show exactly where funds are used and what physical items or results have been produced. Teachers will receive salaries on time, and financial leakage will be reduced. Planners will see where classrooms or laboratories are most needed. Parents and citizens will gain renewed confidence in the city's transparency.

The cumulative effect will be better educational governance. The ERP will allow Kathmandu Metropolitan City to operate with the same efficiency as large national education systems found around the world, while respecting local context and policy requirements.

12. Monitoring, Evaluation, and Continuous Improvement

A formal monitoring and evaluation framework will accompany the system. Indicators such as percentage of schools with fully updated profiles, payroll processing time, investment utilization rates, and data accuracy will be monitored every quarter. Reports will show trends and help adjust operations. The city will establish a steering committee that meets periodically to review progress and authorize upgrades. User satisfaction surveys and technical audits will provide feedback for continuous enhancement.

13. Risk Management and Mitigation

The project recognizes possible risks such as insufficient user adoption, technical glitches, budget constraints, or internet connectivity issues. Mitigation measures include early training, clear documentation, stable hosting, and offline data options. A small contingency fund will be reserved for emergent technical needs. Regular communication with national agencies will minimize policy or integration delays.

ERP Requirements – Stage 1 (Government Schools)

Kathmandu Metropolitan City Education Management ERP

School Profile

Identity Information

- Unique school code linked with EMIS
- Official name of the school
- School level (primary, lower secondary, secondary, or higher secondary)

Establishment and Legal Information

- Year of establishment
- Ward number and location (with potential GIS mapping)
- Accreditation and affiliation documents

Land and Infrastructure

- Total land area and ownership status
- Number and condition of buildings, classrooms, and blocks
- Facility records for library, science lab, ICT lab, sanitation, drinking water, and playground
- Accessibility features for students with disabilities
- Maintenance register with photographic evidence of repairs and renovations

Academic Levels and Streams

- Range of grades offered and records of upgrading history
- Academic streams offered (Science, Management, Humanities, etc.)
- Medium of instruction (Nepali, English, or both)

Governance and Plans

- Principal and Headmaster information with service timeline
- School Management Committee details (chairperson, members, tenure dates)
- School Improvement Plan objectives and annual targets

Enrollment and Class Information

- Total student enrollment by grade, gender, and disadvantaged group
- Number of sections per grade
- Calculated student–teacher ratio

3.2 Staffing and Quota (Darbandi)

- Entry of all sanctioned teacher posts by subject and level
- Identification of teachers appointed under national quota and locally funded positions
- Vacant, filled, and surplus post tracking
- Details of administrative and support staff
- Alerts for posting gaps or surplus teachers

3.3 Payroll and Salary Management

- Salary scales, grades, steps, allowances, and deductions entered once according to national rules
- Recording of basic pay, allowances, tax, gratuity, and provident fund deductions
- Attendance confirmation through manual entry or biometric devices before pay calculation
- Automatic monthly salary computation for teachers and staff
- Review workflow by school head and ward officers, approval by Education Department and Finance Division
- Electronic integration with Metropolitan Financial Management System for direct bank disbursement
- Generation of individual payslips and annual statements for audit and pension records
- Funding source breakdown showing central versus KMC contribution
- Preventive controls against payment beyond sanctioned posts
- Predictive budgeting analysis to forecast salary needs for future fiscal years

3.4 Budget, Investment, and Resource Tracking

- Record of annual budgets at metropolitan city and ward level linked to schools
- Listing of project categories such as Infrastructure, ICT, Green School, Sanitation, Book Free Friday, and Mid-Day Meal
- Entry and digital approval of fund releases with system-based authorization numbers
- Schools upload expenditure records with receipts, photos, and summary notes
- Automatic link of each investment to corresponding school's profile page
- Timeline display showing year-wise and source-wise investments
- Inventory management section for durable assets and consumables
- Reports on budget utilization and percentage of completion

3.5 Integration with National EMIS and External Systems

- Compatibility with national EMIS through secure API or import/export mechanism if possible otherwise manual import
- Periodic data synchronization for school codes, teacher records, and enrollment numbers
- Maintenance of unique IDs to ensure data consistency across systems

- Capacity to submit annual EMIS reports automatically in prescribed format

3.6 Attendance, Examination, and Learning Analytics (Future-Ready Design)

- Database provision for student attendance and performance records
- Analytical tools to relate investments to learning outcomes
- Dashboard indicators for academic trend analysis in future upgrades

3.7 Communication and Citizen Engagement

- Messaging and notification tools for circulars and alerts
- Future provision for parent portal to share school news and events
- Enhancement of parent-school communication and public transparency

3.8 Book Free Friday and Vocational Learning

- Digital module to plan and monitor the Book Free Friday initiative, which replaces textbook teaching with practical learning every Friday
- School planning section to list vocational streams and clubs such as gardening, arts, computer skills, engineering, health care, and environmental activities
- Student registration in chosen clubs and assignment of supervising teachers within the system
- Weekly attendance and activity entry with optional photos and short project summaries
- Ward and Education Department monitoring of participation and progress
- Resource tracking of materials and funds used, linked to budget records
- Automatic connection of Book Free Friday data to student and teacher profiles
- Generation of reports on participation levels, gender balance, activity types, and overall impact